Silas Bambil

Statistics and Quantitative Risk

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Education

Georgia State University **Expected Spring 2028**
MS Mathematics, concentration in Statistics *Atlanta, Georgia*

Georgia State University **May 2025**
MS Quantitative Risk Analysis and Management *Atlanta, Georgia*

Kwame Nkrumah University of Science and Technology **July 2020**
BSc Actuarial Science *Kumasi, Ghana*

Research Appointment

Georgia State University **Aug 2023–Jun 2025**
Graduate Assistant *Atlanta, Georgia*

- Cleaned applicant records and developed predictors for the **Greenberg Sprint** graduate-enrollment model.
- Supported enrollment-model evaluation and provided statistical analysis and data management for faculty risk projects.

Research and Working Manuscripts

- *Selection Regret in Credit-Risk Model Updating Under Distribution Shift: The Role of Recent Labeled Information*. Working manuscript, 2026. Examines how recent labeled-sample size affects cross-validation-based policy selection and predictive regret.
- *Predictive Stability and Adaptation Under Distribution Shift: Calibration, Discrimination, and Decision Thresholds in Credit-Risk Modeling*. With Shudeshna Nandy. Unpublished manuscript, 2026. Compares calibration and updating strategies through simulation and mortgage data.
- *Stable Fire-Loss Quantiles and Far-Tail Expected Shortfall Gaps*. Unpublished manuscript, 2026. Examines tail-risk estimates and model discrepancies in NFIRS fire-loss records.
- *Application of Extreme Value Theory to Fire Insurance*. Undergraduate thesis, KNUST, 2020.

Teaching Experience

Georgia State University **Aug 2026–Present**
Graduate Lab Assistant *Atlanta, Georgia*

- Teach and support College Algebra (MATH 1111) and Support for College Algebra (MATH 0999).

Osceola County School for the Arts **Jan–Jun 2026**
AP Mathematics Teacher *Osceola County, Florida*

- Taught AP Calculus, AP Precalculus, and AP Statistics; students achieved a **70% AP exam pass rate**.

Technical Skills

Software: R, Python, Microsoft Excel.
Methods: Regression, Monte Carlo simulation, bootstrap, calibration, model validation, extreme value theory.
Risk analysis: Credit risk, Value-at-Risk, Expected Shortfall, stress testing, scenario analysis.

Teaching and Research Skills

Teaching: Classroom management; lesson planning in mathematics and statistics; worked-example instruction; assignment and assessment design; student feedback and examination preparation.
Research design: Research-question formulation; literature review; methodological and conceptual framework development; Monte Carlo simulation design.
Analysis and validation: Data preparation; model comparison; separate fitting and evaluation samples; calibration assessment; uncertainty and sensitivity analysis.
Communication: Manuscript and report writing; statistical figures; research presentations; interpretation within stated assumptions and limitations.

Academic Awards

- Graduate Merit Scholarship, Department of Mathematics and Statistics, Georgia State University, August 2026.
- Graduate Merit Scholarship, Robinson College of Business, Georgia State University, August 2023–May 2025.